DDD-M-156d
October 21, 1964

SUPERSEDING
Int. Fed. Spec. DDD-M-00156c (GSA-FSS)
April 24, 1961 and
Fed. Spec. DDD-M-156a
June 7, 1939

FEDERAL SPECIFICATION

MAT, FLOOR, (COIR FIBER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type and three sizes of coir fiber floor mats.

1.2 Classification.

1.2.1 *Type and sizes.* The floor mats covered by this specification shall be of a brush pile type and of the following minimum sizes (see 6.2).

Size 1—30 x 48 inches (no. 9 commercial designation).
Size 2—22 x 36 inches (no. 5 commercial designation).
Size 3—18 x 30 inches (no 3 commercial designation).

NOTE Measurements are to be made on the reverse side of the mat.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276—Thread, Cotton.
QQ-S-781—Steel, Strapping, Flat.
UU-P-271—Paper, Wrapping, Waterproofed Kraft.
CCC-T-191—Textile Test Methods.
DDD-L-20—Label; for Clothing, Equipage, and Tentage, (General Use).
PPP-P-291—Paperboard, Wrapping, Cushioning.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed Std No. 123—Marking for Shipment and Storage (Civilian Agencies).
Fed. Std. No 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S Government Printing Office, Washington, D C 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.

Military Specification:

MIL-T-7807—Thread, Nylon.

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Military Standards:
MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Textile Fiber Products Identification Act

(Copies may be obtained, free of charge, from the Federal Trade Commission, Washington, D C 20407.)

3 REQUIREMENTS

3.1 **Material.**

3.1.1 *Coir fiber.* Coir fiber used for making coir yarns for mats covered by this specification shall be from the cocoanut husk and shall be of natural color.

3.1.2 *Tufting yarn.* The tufting yarn (brush) shall be of vycome type medium weaving or fine weaving coir yarn, and each tuft shall consist of not less than 3 laid yarns each consisting of not less than 2 ply yarn with adequate twist to meet the additional requirements contained herein The average breaking strength of final plied yarn shall be not less than 25 pounds when tested as specified in 4.1.2 A certificate of compliance stating that the tufting yarn is vycome type medium weaving or fine weaving coir yarn shall be submitted with each lot. An identified representative twelve foot section of the tufting yarn shall be submitted with each lot

3.1.3 *Warp back yarn.* The warp back yarn shall be not less than 2 ply coir yarn of a quality to meet the requirements of this specification. The average breaking strength of final plied yarn shall be not less than 35 pounds when tested as specified in 4.1.2. An identified representative twelve foot section of the warp back yarn shall be submitted with each lot.

3.1.4 *Filling yarn.* The filling yarn shall be not less than 2 ply coir yarn of a quality to meet the requirements of this specification. The average breaking strength of final plied yarn shall be not less than 30 pounds when tested as specified in 4.1.2 An identified representative twelve foot section of the filling yarn shall be submitted with each lot.

3.1.5 *Braid.* The braid for binding the edges of the mat shall be made from not less than 7 coir yarns, each yarn of not less than 2 ply vycome medium weaving, fine weaving, or anjengo hard twist yarn The width of the finished braid shall be not less than 5/8 inch. The average breaking strength of the ply yarns shall be not less than 30 pounds when tested as specified in 4.1.2. A certificate of compliance stating that the braid yarns are vycome medium weaving, fine weaving or anjengo coir yarns shall be submitted with each lot. An identified representative twelve foot section of the braid yarn shall be submitted with each lot

3.1.6 *Sewing material (for braid).* Sewing material for attaching the coir braid binding to the mat shall be either cotton sewing thread, 3.1.6.1; nylon sewing thread, 3.1.6.2, or coir yarn 3.1.6.3; at the option of the manufacturer.

3.1.6.1 *Cotton* Cotton sewing thread shall be any shade of brown color, unbleached, or white, and shall conform to the requirements of type IVB, 7 ply, ticket no. 8/7 of V-T-276 There are no colorfastness requirements for colored thread used for mats covered by this specification.

3.1.6.2 *Nylon.* Nylon sewing thread shall be any shade of brown color or white, type

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I or II, class 1, No. 4 cord of MIL-T-7807. There are no colorfastness requirements for colored thread used for mats covered by this specification.

3.1.6.3 *Coir yarn.* Coir yarn for sewing the braid shall be of 2 ply medium twist anjengo quality or vycome quality having an average breaking strength of not less than 32 pounds when tested as specified in 4.1.2. A certificate of compliance stating that coir yarn is anjengo or vycome quality shall be submitted with each lot. An identified representative twelve foot section of the coir yarn shall be submitted with each lot.

3.2 **Construction.** The mat shall be fabricated from the tufting yarns, warp back yarns and filling yarns specified in 3.1.2, 3.1.3 and 3.1.4. The weave shall be of the diamond or straight carpet back construction. The height of the tufts (when measured from the back of the mat to the top of the tufts) shall be not less than 1-1/2 inches. The mat shall consist of not less than 1-1/2 tufting yarns, warp back yarns and filling yarns per inch when tested as specified in 4.3. All sides of the mat shall have a continuous braid, specified in 3.1.5, securely attached with sewing material conforming to 3.1.6 The ends of the braid shall be overlapped for a length of not less than 3/4 inch and fastened together with twine, thread or rust resistant metal. There shall be not less than a average of 2-1/2 stitches per inch using stitch type 301 or 401 of Fed. Std. No 751 when using cotton or nylon sewing thread, and not less than 1 stitch per inch when using coir yarn.

3.3 **Color** The color of the mat shall be a natural tan, of uniform shade, reasonably free from streaks, and shall fall within an established shade range mutually agreed upon by the contracting officer and the supplier (see 6.2).

3.4 **Weight.** The mats shall be of the following minimum weights:

Size 1—16 pounds
Size 2—9 pounds
Size 3—6 pounds

3.5 **Marking for identification.** Each mat shall have a label conforming to type I, class 9 of DDD-L-20 securely attached to the back of the mat. The label shall contain the following information printed in bold letters and numbers not less than 1/8 inch in height:

Nomenclature
Size Federal Stock Number
Contractor's name or trademark
Contract number and date

3.6 **Fiber identification.** Each mat shall be labeled, ticketed, or invoiced for fiber content in accordance with the rules and regulations under the Textile Fiber Products Identification Act

3.7 **Workmanship.** The finished mat shall conform to the quality of product established by this specification and the face of the mat shall be free of loose tufts, uneven tufts and reasonably free from streaks. The occurrence of defects shall not exceed the specified acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier's responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Component inspection.* In accordance with 4.1, the supplier is responsible

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for insuring that components used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.1.2 *Testing of components.* In addition to the quality assurance testing provision of the subsidiary specifications, testing of the components shall be performed in accordance with table I for the characteristics shown therein. The values specified in section 3 of this specification apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method The lot size shall be expressed in units applicable to the component being tested, i.e., tubes, cones, spools, skeins, hanks, etc. The sample unit shall be twelve linear feet. When the sample size for components covered by the subsidiary specifications is not specified, the sample size for each characteristic shall be in accordance with inspection level S-2 of MIL-STD-105 except that for lot sizes of 501 (units of product) and over the sample size shall be 5 The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified.

TABLE I—*Instructions for testing components*

Component	Characteristic	Requirement paragraph	Test method
Tufting yarn (coir)	Breaking strength[1]	3.1.2	4100[2] CCC-T-191
	Material identification	3.1.2	1270 or 1271 CCC-T-191
Warp back yarn (coir)	Breaking strength[1]	3.1.3	4100[2] CCC-T-191
	Material identification	3.1.3	1270 or 1271 CCC-T-191
Filling yarn (coir)	Breaking strength[1]	3.1.4	4100[2] CCC-T-191
	Material identification	3.1.4	1270 or 1271 CCC-T-191
Braid (coir)	Breaking strength[1]	3.1.5	4100[2] CCC-T-191
	Material identification	3.1.5	1270 or 1271 CCC-T-191
Sewing material (coir)	Breaking strength[1]	3.1.6.3	4100[2] CCC-T-191
	Material identification	3.1.6.3	1270 or 1271 CCC-T-191

[1] The breaking strength of any individual specimen shall be not less than 80 percent of the specified minimum breaking strength of the sample unit average.

[2] Ten specimens shall be tested in lieu of twenty specimens as required in method 4100.

4.2 Examination of the end item

4.2.1 *Visual examination.* The lot size shall be expressed in units of one mat each. The inspection level for purposes of determining the sample size shall be II of MIL-STD-105. The unit of product shall be one mat. The defects found during the examinations shall be classified in accordance with table II. The acceptable quality level (AQL) shall be for major defects and 4.0 for total defects (major and minor combined) per hundred units.

TABLE II—*End item defects*

Defect	Classification	
	Major	Minor
Tufting yarn not as specified	X	
Warp back yarn not as specified	X	
Filling yarn not as specified	X	
Braid not as specified	X	
Sewing material not as specified		X
Weave other than diamond or straight carpet back	X	
Binding sewed with cotton or nylon thread		
Less than 2-1/2 stitches per inch	X	
Color of thread not as specified		X
Other than stitch type 301 or 401	X	
Binding sewed with coir yarn		
Less than 1 stitch per inch	X	
Excessively mottled or streaked on face of mat	X	
Uneven pile (tufts) on face		X
Top of tufts excessively twisted		X

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Defect	Classification	
	Major	Minor
Loose tufts, not secured by weave	X	
Tufts not completely covering base weave	X	
Color other than specified	X	
Identification label—missing, incorrect, text incomplete, illegible or insecurely attached		X

4.2.2 *Dimensional examination.* The lot size shall be expressed in units of one mat each. The inspection level for purposes of determining the sample size shall be S-2 of MIL-STD-105. The unit of product shall be one mat. The defects found during the examinations shall be classified in accordance with table III. The acceptable quality level (AQL) shall be 4.0 defects per hundred units.

TABLE III—*Dimensional defects*

Examine	Defect
Size*	Less than specified
Height of tufts	Less than 1-1/2 inches
Width of binding	Less than 5/8 inch
Overlapped ends of binding	Less than 3/4 inch

* Size measurements of mats shall be made on the reverse (floor) side of the mat.

4.2.3 *Weight examination.* The lot size shall be expressed in units of one mat each. The inspection level for purposes of determining the sample size shall be S-2 of MIL-STD-105. The unit of product shall be one mat. The acceptable quality level (AQL) shall be 4.0 defects per hundred units.

TABLE IV—*Weight defects*

Examine	Defect
Weight	Less than specified

4.2.4 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table V. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per 100 units.

TABLE V—*Preparation for delivery defects*

Examine	Defects
Marking	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as: Incomplete closure of container flaps, improper taping or loose strapping. Bulging or distortion of containers.
Content	Number per container is more or less than specified.
Packing	Not level specified. Container material or construction not as specified. Closures not made by specified or required methods or materials

4.3 **Testing of the end item.** The method of testing specified in CCC-T-191 as listed in table VI shall be followed. The values specified in section 3 of this specification apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The lot size shall be expressed in units of mats each. The sample unit shall be one finished mat each. The sample size for each characteristic shall be in accordance with inspection level S-2 of MIL-STD-105 except that for lot sizes of 501 mats and over the sample size shall be 5. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified.

TABLE VI—*Instructions for testing end item*

Characteristic	Requirement reference	Test method
Yarns per inch	3.2	5050

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5. PREPARATION FOR DELIVERY

(For civil agency procurement, Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5 1 Packaging. Not applicable.

5.2 Packing. Packing shall be level A, B, or C as specified, (see 6.2).

5.2.1 *Level A* Packing shall be in accordance with 5.2.2, except when specified, waterproof kraft wrapping paper meeting the requirements of class B-2 of UU-P-271, shall be used as an overwrap and waterproof sealed in accordance with good commercial practice

5.2.2 *Level B* Four (4) mats size 30 x 48 inches, or six (6) mats size 22 by 36 inches, or twelve (12) mats size 18 by 30 inches shall be stacked flat, with all edges even, and wrapped in paperboard (with ends left open) conforming to type III, class 2, of PPP-P-291 The wrapped bundle shall be secured by three evenly spaced 3/8 inch minimum width steel straps encircled girthwise conforming to type I, any class or grade, of QQ-S-781 Use of plastic laminated nonwoven tape made of natural or synthetic yarns having the equivalent strength of steel straps is permissible in lieu of steel straps.

5.2.3 *Level C.* Mats shall be packed in a manner that will insure acceptance by common carrier and safe delivery at destination Shipments shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3 1 *Civil agencies* In addition to any special marking specified in the contract or order, each shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military.* In addition to any special marking specified in the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use Mats covered by this specification are intended for use in doorways to clean soles of footwear.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Size (1.2.1).
(*c*) Color (3.3).
(*d*) Selection of applicable levels of packing (5.2)
(*e*) Packing with waterproof paper (5.2.1)

6.3 Transportation description The transportation descriptions and minimum weights applicable to this commodity are:

Rail·

Mats, fiber, not otherwise indexed by name, other than cotton, rayon, synthetic fiber, silk or wool.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor

Mats, coir fiber
Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.4 Level B packing shall be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

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